

Md. Asiful Islam Miah

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Google Scholar · GitHub · LinkedIn · Portfolio

RESEARCH SUMMARY

I am a researcher in computer vision and medical AI, focused on efficient deep learning architectures for 3D spatial understanding and clinical image analysis. My work spans chest X-ray disease detection and localization, brain age estimation from T1-weighted MRI, and differentiable 3D voxel tokenization for spatial visual question answering. I aim to develop AI systems that are computationally efficient and clinically deployable — bridging frontier deep learning methodology with real-world medical imaging applications.

RESEARCH INTERESTS

Computer Vision · Medical Image Analysis · Vision-Language Models · 3D Deep Learning · Efficient Neural Architectures · Spatial Reasoning · Robotics Perception

EDUCATION

Khulna University of Engineering & Technology (KUET)

B.Sc. (Eng.) in Computer Science and Engineering

Jan 2018 – Feb 2023

Khulna, Bangladesh

CGPA: 3.77 / 4.00 Class Rank: 13 / 127 (Top 10%)

Undergraduate Thesis: “InfLocNet: Enhanced Lung Infection Localization and Disease Detection from Chest X-Ray Images Using Lightweight Deep Learning”

Supervisor: Prof. Dr. M. M. A. Hashem, Dept. of CSE, KUET

- Dean’s Award List — 2019, 2020, 2021 (awarded to top academic performers each year)
- Technical Scholarship — 2018–2021, KUET (merit-based award for academic excellence)
- President, HACK — Hardware Acceleration Club of KUET (organised Robotics & embedded systems workshops for 100+ members)
- Member, SGIPC — Special Group of Interest in Programming Contest, KUET

PUBLICATIONS

*Lead / corresponding author. Co-author affiliations in parentheses.

[1] GeoToken: Differentiable 3D Voxel Tokenization for Geometric Spatial Reasoning in VLMs.

Md. Asiful Islam Miah* (Southeast University), Md. Nazrul Islam (Southeast University), S M Taslim Uddin Raju (University of Waterloo), Hamdi Altaheri (King Saud University)

Under review — AAAI 2027 [Preprint available on request] [Top-tier AI venue – Lead author]

[2] InfLocNet: Enhanced Lung Infection Localization and Disease Detection from Chest X-Ray Images Using Lightweight Deep Learning.

Md. Asiful Islam Miah* (KUET), Shourin Paul (KUET), Sunanda Das (KUET), M. M. A. Hashem (KUET)

In preparation — Target: IEEE Transactions on Medical Imaging, 2026

Preprint : arXiv:2408.06459 [In preparation – Lead author – Extends undergraduate thesis]

[3] Predicting Brain Age Using Lightweight 3D CNN Architecture from T1-Weighted MRI Images.

Md. Asiful Islam Miah* (KUET), Shourin Paul (KUET), Md. Ahsan Habib (KUET), Kazi Saeed Alam (KUET)

Springer Nature Singapore, Lecture Notes in Electrical Engineering (Scopus indexed) — ICEEE 2024

doi:10.1007/978-981-97-9112-5_9 [Published – Lead author]

[4] Chest-InfNet: A Deep Learning Architecture for Lung Diseases Detection and Infected Region Localization from Chest X-Ray Images.

Shourin Paul (KUET), Md. Asiful Islam Miah (KUET), Sunanda Das (KUET), M. M. A. Hashem (KUET)

PROFESSIONAL EXPERIENCE

Southeast University

Lecturer, Department of Computer Science & Engineering

Nov 2023 – Present
Dhaka, Bangladesh

- Teaching CSE-4101 (Deep Learning) and CSE-3201 (Computer Vision) to 60+ undergraduate students per semester
- Supervising 3 undergraduate thesis projects in medical image segmentation and automated disease classification from radiology images
- Conducting independent research in 3D visual understanding and medical AI in parallel with full teaching responsibilities, resulting in an active AAAI 2027 submission
- Established a weekly AI reading group focused on vision-language models and efficient inference methods (2024–present)

Shanghai Baud Communication Technology Co., Ltd. (BDCOM)

R&D Engineer

China-based multinational networking equipment company

Mar – Oct 2023
Dhaka, Bangladesh

- Designed and implemented Layer 2/3 networking protocols (RSTP, STP, OSPF) in C/C++, integrated into a real-time operating system (RTOS) kernel for enterprise-grade managed network switches
- Collaborated with the firmware team on protocol stack validation and regression testing for production firmware deployed across commercial switch product lines
- Gained systems-level expertise in embedded Linux and bare-metal RTOS environments, complementing deep learning research with low-level programming depth

TECHNICAL SKILLS

Deep Learning & CV

PyTorch, TensorFlow/Keras, OpenCV, 3D CNNs, Vision Transformers (ViT), Vision-Language Models, Object Detection, Image Segmentation, Weakly Supervised Learning

Medical Imaging

NIfTI, DICOM, SimpleITK, 3D MRI Analysis, Chest X-Ray Processing, Lesion Localisation, FreeSurfer (familiarity)

Languages & Tools

Python (primary), C/C++ (systems), CUDA (basic), Git, Linux, NumPy, scikit-learn, Matplotlib, Weights & Biases

Human Languages

Bengali (native), English (proficient)

RESEARCH PROJECTS

GeoToken — 3D Spatial Visual Question Answering

2025 – Present

- Developed a differentiable 3D voxel tokenization pipeline enabling efficient grounded reasoning over 3D scenes; novel tokenisation reduces computational overhead versus dense point-cloud methods
- First-author submission to AAAI 2027 (~20% acceptance rate); preprint available on request

InfLocNet & Chest-InfNet — Lung Disease Detection from Chest X-Ray

2022 – Present

- Designed lightweight CNN architectures for multi-class lung disease classification and weakly supervised infected-region localisation from chest X-ray images without pixel-level annotation
- Chest-InfNet: Scopus-indexed Taylor & Francis book chapter, BIM 2023 (doi:10.1201/9781032650715-6)
- InfLocNet: undergraduate thesis work, in preparation for *IEEE Transactions on Medical Imaging* (2026)

- Proposed a lightweight 3D CNN achieving competitive brain age estimation accuracy with significantly fewer parameters than prior state-of-the-art, enabling deployment on resource-constrained hardware
- Published in Springer Nature LNEE (Scopus indexed), ICEEE 2024 — doi:10.1007/978-981-97-9112-5_9

AWARDS & HONOURS

- Dean's Award List — KUET (2018, 2019, 2020, 2021)
- Technical Scholarship — KUET (2017–2021), merit-based award for academic excellence
- Intra KUET Programming Contest — 2019, 2020
- Programming Contest, BitFest KUET IUPC — 2019
- Line Follower Robot Contest — Ignition, KUET, 2018

LEADERSHIP & ACADEMIC SERVICE

- **President**, HACK — Hardware Acceleration Club, KUET: led Robotics and embedded systems workshops for 100+ student members
- **Member**, SGIPC — Special Group of Interest in Programming Contest, KUET: competitive programming training and inter-university contest participation
- **AI Reading Group Coordinator**, Southeast University: weekly seminars on vision-language models and efficient deep learning (2024–present)

REFERENCES

Prof. Dr. M. M. A. Hashem

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